IT IMPLEMENTATION ON LIBRARY

Group 10

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CEP CCIT

FAKULTAS TEKNIK UNIVERSITAS INDONESIA

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PROJECT ON

IT Implementation on Library

Developed by

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3. **Muhammad Adib Arkan**



IT Implementation on Library

Batch Code : 1CC1

Start Date : September 19, 2024

End Date : November 2, 2024

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Date of Submission: November 2, 2024

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CERTIFICATE

This is to certify that this report titled “Library” embodies the original work done by Laura Al Khansa, Namira Assalwa and Muhammad Adib Arkan. Project in partial fulfillment of their course requirement at NIIT.

Coordinator:
Tri Agus Riyadi, S. Kom, MT

ACKNOWLEDGEMENT

First of all, we would like to thank to God because of His blessings and guidance, author can compose and present a paper that contains about "**IT Implementation on Library**". The author also thanked to Mr. Tri Agus Riyadi as lecturer that have provided guidance to the author in process of preparing this paper. The author realizes that this project still have many shortcomings and far from being perfect. Therefore, the authors expect constructive criticism and suggestion to improve this paper and can be reference in preparing the papers or subsequent tasks. The authors also apologize if in writing this paper there are typos and errors that confuse the reader in understanding the author's intent.

SYSTEM ANALYSIS

System Summary: The library is an institution that provides various sources of information, including books, journals, and digital access at affordable prices, making it one of the favorite places for students and students to study and interact with friends. The library is equipped with computer and wifi facilities, allowing visitors to work on college assignments, access online materials, and utilize digital library services. With a comfortable and quiet atmosphere, the library is also an ideal place to create content on social media while exploring the world of knowledge.

System Process: The implementation of Local Area Network (LAN) in the library allows the integration of various devices and systems such as CCTV, PC, borrowing and returning machines. With LAN, the library can connect these devices efficiently, facilitate visitors to borrow and return books independently through integrated machines, and monitor security through connected CCTV. In addition to LAN, the library can also use cloud storage to store data such as CCTV footage and borrowing transaction history, reducing the need for large local storage devices. This technology supports more efficient resource management, faster information access, and better security for all visitors.

Hardware

1. Smart Pustakawan Buku Detetor Library Management ISO15693 Library Reader



Figure 1. HC-LB693 (Ref: https://id.made-in-china.com/co_hecere/product_Smart-Libraray-Books-Detetor-RFID-Library-Management-ISO15693-Library-Reader_ysyueshyhg.html)

Basic Specification:

Protocol	ISO15693, ISO1800-3 OR EPC GEN2 ISO18000-6C
Display	15.6-inch touchscreen.
Resulution	1920*1280
Operating Systems	Windows 7, 0/10.0 or Android.
Industrial Computer	Intel Celeron J1900 2.0GHz
Active Frequency	13.56 MHz or 860-930 MHz
Communication Port	RJ45, USB, or WIFI.
Size	550*520*402mm

2. Smart library automatic borrow and return books Touch Screen RFID card



Figure 2. OEM/ODM (Ref: https://www.alibaba.com/product-detail/Smart-library-automatic-borrow-and-return_60840144193.html?spm=a2700.details.you_may_like.2.2ec64af4cGCs1Y**)**

Basic Specification:

Installation	Floor standing
Material	Aluminum Case+Tempered Glass Panel
Type	LCD Touch Screen
Operating Systems	Android/Window 7/8/10(optional)

3. Hikvision Turbo EXIR ECO Series Camera DS-2CE78H0T-IT1F Turbo 5MP, 8MP



Figure 3. DS-2CE78H0T-IT1F (Ref: https://www.monotaro.id/s041695975.html**)**

Basic Specification:

Model Number	DS-2CE78H0T-IT1F
Horizontal Resolution	5MP
Form Factor	Eyeball
Lens	Fixed Lens
Power Supply (V)	12 DC

4. Anti Theft Alarm Range RS232 Range RS232 RS485 RFID Reader



Figure 4. HC-TDM02 (Ref: https://indonesian.alibaba.com/product-detail/RFID-Security-Gate-UHF-Fixed-860-1601042275600.html l**)**

Basic Specification:

Frequency	AS: 902-928 MHz; UE: 865-868 MHz
Protocol	ISO18000-6C (EPC GEN2 CLASS ONE), ISO 18000-6B
Amplification	7dBi
Operating Mode	Fixed Frequency or Software Control
RF Power	0-30dBm, Software Control and adjustable
Work Mode	Passive and active for reading, can be controlled with software.
Communication Port	Wiegand 26/34/98, RS232, RS485, TCP/IP
Input Interface	One-way trigger input
Reading range	1-4M
Work Model	Green LED, Red LED if Alarm
Catu Daya	DC+220v+-10%
Size	380*62*1440mm

5. Epson Printer Colour Multi Function L Series L3250



Figure 5. L3250 (Ref: https://www.monotaro.id/s032817102.html)

Basic Specification:

Dimensions (WxDxH) (mm)	375x347x179
Maximum Resolution (dpi)	5760x1440
Nozzle Configuration	180x1 nozzles Black, 59x1 nozzles per colour (Cyan, Magenta, Yellow)
Scanner Type	Flatbed colour image scanner
Operating System Compatibility	Windows XP / Vista / 7 / 8 / 8.1 / 10, Windows Server 2003 / 2003 x64 / 2008 / 2012 / 2016, Only printing and scanning functions are supported for Windows Server OS, Mac OS X 10.6.8 or later
Connectivity	USB 2.0, WIFI DIRECT
Printer Language	ESC/P-R, ESC/P Raster
Power Consumption (W)	Operating: 12; Sleep: 0.7; Power Off: 0.2; Standby: 4.5

6. Panasonic AC Single Split Non-inverter CS-PN18TKP 2PK



Figure 6. CS-PN18TKP (Ref: https://www.monotaro.id/s012172474.html)

Basic Spesification:

Cooling Capacity (kW)	5.28
Dehumidification (L/h)	2.9
Air Circulation (m^3/min)	19.7 / 695 (indoor)
Noise Level (dB-A)	indoor: 45/39, outdoor: 54
Dimensions W x D x H (mm)	1102 x 244 x 302
Voltage (V)	220
Capacity (PK)	2
Cooling Capacity (BTU/h)	18000

7. HP PC Computer OS Windows 10 Home + OHS 2019 24-df1042d AiO White RAM 8GB



Figure 7. 24-df1042d AiO (Ref: https://www.monotaro.id/s039535962.html)

Basic Spesification:

Processor	Intel Core i5-1135G7 Processor (2.4 GHz; 8M Cache; up to 4.2 GHz)
RAM	8GB
Storage	512GB SSD
Screen Type	23.8 Inch FHD (1920×1080), Touch Screen
Detail Graphics	Intel Iris Xe Graphics
Connectivity	1xAudio Jack, 1xCardReader, 1xHDMI, 1xKensington Lock, 1xRJ45, 2x USB 3.0, 2xUSB 2.0
Networking	Bluetooth 4.2, WiFi

8. Modem Router Wireless Wifi MyRepublic 2744R 3G



Figure 8. Modem router (Ref: https://shopee.co.id/drisdianto/15558084404)

Basic Spesification:

Antenna Type	Internal
Modem Type	Cable
Number of LAN Ports	4
Wireless Network	3G, Wifi N (802.11n)
Warranty Type	Manufacturer Warranty
Modem Connection Speed	150Mbps

9. tp-link 24Port Gb L2 Managed Switch with 4 SFP Slots TL-SG3428



Figure 9. Switch (Ref: https://www.monotaro.id/s030842175.html)

Basic Spesification:

Interface	24× 10/100/1000 Mbps RJ45 Ports• 4× Gigabit SFP Slots• 1× RJ45 Console Port• 1× Micro-USB Console Port
Environment	Operating Temperature (°C): 0-45 (32-113°F)Storage Temperature (°C): -40 to 70 (-40 to 158°F)Operating Humidity: 10-90% RH non-condensingStorage Humidity: 5-90% RH non-condensing
Dimensions (WxDxH) (inch)	17.3x7.1x1.7 (440x180x44mm)
Max Power Consumption (W)	19.22 (220V/50Hz)
Max Heat Dissipation (BTU/h)	65.58 (110V/60Hz)
Packet Forwarding Rate (Mpps)	41.66
Power Supply (V/Hz)	100-240 AC/50-60
Mounting	Rack Mountable

10. BELDEN Kabel UTP (Kabel LAN) 7814A 008 Grey 305m 1roll(305m)

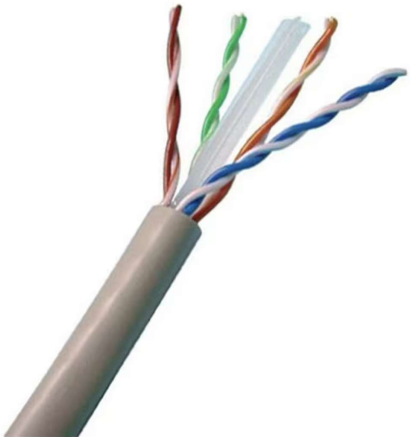


Figure 10. Straight (Ref: https://www.monotaro.id/s011464730.html)

Basic Spesification:

Category	6
No. of Pairs	4
Conductor Size	24
Nominal Diameter (mm)	5.8
Jacket	PVC
Insulation	Polyolefin
Suitability	Indoor
Standardization	TIA568C.2

11. Vention Kabel HDMI Male to Male 4K Performance Series AAH 8m



Figure 11. HDMI (Ref: https://www.monotaro.id/s031075169.html)

Basic Spesification:

AWG	28AWG
Connector Shell	PVC
Connector Interface	Standard HDMI Male to Male
Connector Material	Gold-plated
Conductor Material & Shielding	Pure Copper & Double Aluminium Foil + Metal Braided
Jacket	PVC

12. CAROL Coaxial Cable C57753101 18 AWG Conductor Gauge, 1000 ft Cable Lg Rg-6/U Black



Figure 10. Carol (Ref: https://www.monotaro.id/s041418093.html)

Basic Spesification:

Standards	ASTM B3; ASTM B869; ETL Listed; UL Listed
Shielding material	100% Flexfoil; 60% Aluminum Braid
NEC Article and Type	725 CL2; 800 CM; 820 CM
Max. Frequency (MHz)	1,000
UNSPSC	26121606
Cable Fire Rating	General
Coaxial Cable Type	RG-6/U
Insulation Material	FPE

Software

1. Koha (Library Management System)

   Koha acts as the main system that manages all aspects of library operations, including book catalogs, borrowing, returning, and member management. This system will make it easier for visitors to search for books, view availability status, and borrow and return automatically through integration with RFID or barcode scanners. In addition, Koha helps library administration track book usage statistics, late fines, and inventory status in real-time.

2. Bibliotheca RFID (RFID/BARCODE System)

   Bibliotheca RFID supports automated book lending and returning machines with RFID technology. Books equipped with RFID tags can be quickly scanned by users at the self-service machines, speeding up the borrowing and returning process. The system also improves the accuracy of book tracking, minimizes human error, and allows users to borrow or return books without having to queue at a manual service desk.

3. Milestone XProtect (Security and Monitoring Software)

   Milestone XProtect acts as an integrated surveillance platform that manages CCTV and body check machines in the library. With motion detection and real-time monitoring, the system allows library staff to monitor the safety and security of visitors and safeguard critical areas from unauthorized access. CCTV footage is automatically stored and can be reviewed for incident investigations, ensuring security throughout the facility.

4. Honeywell BMS (Building Management System)

   Honeywell BMS functions to manage and control the physical environment of the library, including the AC system, lighting, and other energy systems. With this system, the room temperature can be automatically adjusted according to the number of visitors or operating hours, maintaining comfort while saving energy. This BMS can also provide notifications if there is a disruption to devices such as AC or air conditioning systems.

5. PaperCut (Printing Management Software)

   PaperCut plays a role in managing all printing activities in the library. This software ensures that printer usage runs efficiently by managing print queues and providing access control to users based on predetermined permissions. Library users can send documents to be printed, while admins can monitor the number of prints produced to control operational costs and optimal resource usage.

6. Asset Panda (Admin & Inventory Management Software)

   Asset Panda helps libraries efficiently manage their inventory of books, hardware, and other assets. The software enables centralized asset tracking, making it easy to record stock of books, electronic equipment, and administrative supplies. This way, libraries can monitor the condition of items, conduct inventory audits, and minimize the risk of undetected asset loss or damage.

7. HID Global (Access Control System)

   HID Global provides access control solutions to manage who is allowed into specific areas of a library, such as administration or archives. These systems can be integrated with access cards or body check systems to ensure that only individuals with the proper authorization can access those areas. This increases security while maintaining the privacy of sensitive areas within the library.

8. Acronis Backup (Backup and Data Security Software)

   Acronis Backup protects critical library data by providing an automated backup system that keeps loan transaction data, member information, and book inventory secure. In the event of a system failure, data loss, or cyberattack, data can be easily restored from previous backups. This ensures the library's operations continue without major disruptions in information management.

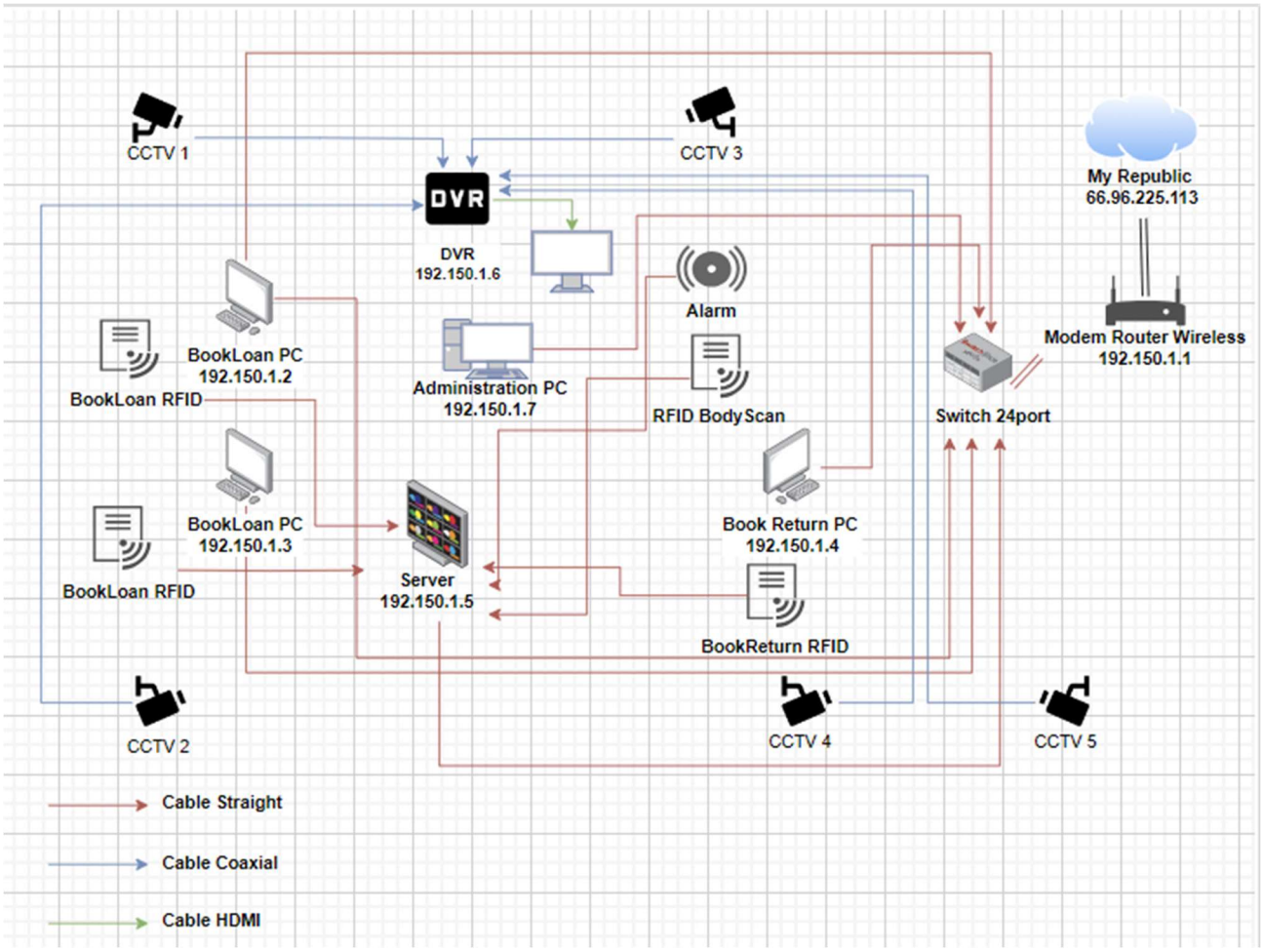
9. Windows (Operating System)

   Windows as the operating system on the administration PC will manage various software used in the library, such as Koha (Library Management System), CCTV monitoring software, and other software. This system also facilitates integration with hardware such as printers, body check machines, and RFID readers. With Windows, library staff can work in an environment that is compatible with various applications and hardware used.

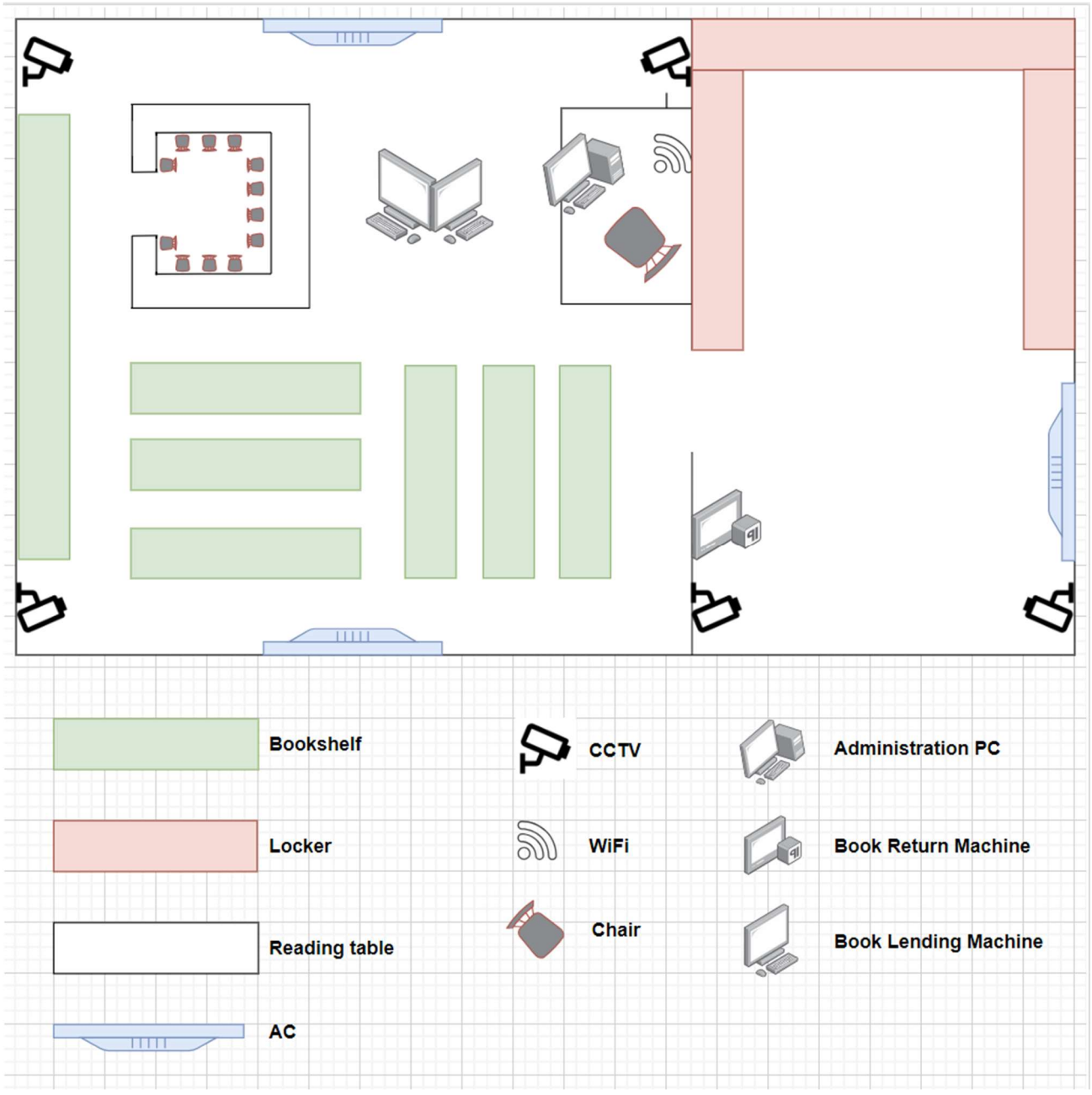
10. Microsoft Office (Productivity Software)

    Microsoft Office is needed to support administrative and documentation tasks in the library. For example, Excel can be used to create book inventory reports, loan data, and visitor statistical analysis. Word is useful for creating documents, official letters, or announcements. PowerPoint can help in presenting library information or staff training, while Outlook can be used for internal or external communication via email.

NETWORK TOPOLOGY DESIGN



LIBRARY'S FLOOR PLAN



COST

Hardware

No	Hardware Specification	Unit	Unir Price	Total Price
1	Smart Pustakawan Buku Detetor Library Management ISO15693 Library Reader	2	Rp19.805.688,-	Rp39.611.376,-
2	Smart library automatic borrow and return books Touch Screen RFID card	1	Rp19.646.958,-	Rp19.646.958,-
3	Hikvision Turbo EXIR ECO Series Camera DS-2CE78H0T-IT1F Turbo 5MP, 8MP	5	Rp799.000,-	Rp3.895.000,-
4	Anti Theft Alarm Range RS232 Range RS232 RS485 RFID Reader	2	Rp11.040.663,-	Rp 22.081.326,-
5	Epson Printer Colour Multi Function L Series L3250 1unit	1	Rp3.099.000,-	Rp3.099.000,-
6	Panasonic AC Single Split Non-inverter CS-PN18TKP 2PK	3	Rp10.999.000,-	Rp32.997.000,-
7	HP PC Computer OS Windows 10 Home + OHS 2019 24-df1042d AiO White RAM 8GB	1	Rp28.999.000,-	Rp28.999.000,-
8	Modem Router Wireless Wifi MyRepublic 2744R 3G	1	Rp150.000,-	Rp150.000,-
9	tp-link 24Port Gb L2 Managed Switch with 4 SFP Slots TL-SG3428	1	Rp3.499.000,-	Rp3.499.000,-
10	BELDEN Kabel UTP (Kabel LAN) 7814A 008 Grey 305m 1roll(305m)	1	Rp3.099.000,-	Rp3.099.000,-
11	Vention Kabel HDMI Male to Male 4K Performance Series AAH 8m	1	Rp209.000,-	Rp209.000,-
12	CAROL Coaxial Cable C57753101 18 AWG Conductor Gauge, 1000 ft Cable Lg Rg-6/U Black	1	Rp15.999.000,-	Rp15.999.000,-
SUB TOTAL HARDWARE				**Rp173.285.660,-**

Software

No	Hardware Specification	Unit	Unir Price	Total Price
1	Koha (per year)	1	Rp10.750.000,-	Rp10.750.000,-
2	Bibliotheca RFID (per tag)	2	Rp151.431.000,-	Rp302.862.000,-
3	Milestone XProtect	1	Rp15.575.000,-	Rp15.575.000,-
4	Honeywell BMS	1	Rp100.750.000,-	Rp100.750.000,-
5	PaperCut	1	Rp15.143.100,-	Rp15.143.100,-
6	Asset Panda	1	Rp30.286.200,-	Rp30.286.200,-
7	HID Global	1	Rp.20.000.000,-	Rp.20.000.000,-
8	Acronis Backup	1	Rp1.000.000,-	Rp1.000.000,-
9	Windows 10 Pro	3	Rp2.500.000,-	Rp7.500.000,-
10	Office 365 Personal	3	Rp2.000.000,-	Rp6.000.000,-
SUB TOTAL SOFTWARE				**Rp509.866.300,-**

SUB TOTAL HARDWARE	Rp173.285.660,-
SUB TOTAL SOFTWARE	Rp509.866.300,-
MY REPUBLIC	Rp. 795.000,-
TOTAL BUDGET	**Rp683.946.960,-**

PROJECT FILE DETAILS		
No	**File Name**	**Remarks**
1	System Analys	Laura Al Khansa, Muhammad Adib
2	Hardware	Laura Al Khansa
3	Software	Laura Al Khansa, Namira Assalwa
4	Topology	Laura Al Khansa
5	Library's Floor Plan	Laura Al Khansa, Namira Assalwa, Muhammad Adib
6	Budget	Laura Al Khansa, Namira Assalwa
7	Document	Laura Al Khansa
8	Powerpoint Presentation	Laura Al Khansa